

Lecture Notes on QFT I

October 28, 2025

Readings

- Peskin & Schroeder Sec. 2.1 and 2.2
- Weinberg vol 1 Chap. 1

Lecture 1: Principle of Locality and Fields

The goals of this Quantum Field Theory class is

- To develop concepts, formalism and techniques for quantum dynamics of fields.
- Quantum Field Theory is also very important for other interactions in nature. Among the four fundamental interactions in nature, three of them are described by QFT completely.
- Even though QFT is initially developed for particle physics, but over the years, QFT has been found many applications in many other branches of physics, in condensed matter, statistical physics and so on. It's fair to say, nowadays, QFT has become a universal language for theoretical physics in all different fields.

Why we consider QFT?

principle of locality

The first important concept is called the **principle of locality**.

In Newtonian mechanics, you have action at a distance, for example, if you look at gravity and the gravity is exerted by the sun on the Earth, but they are very far away. And the same thing with the Coulomb interaction between the charged particles.

But then, in the 19th century, they came from this principle of locality, that is formulated by Faraday around 1830. The principle of locality says that all points you don't have action at a distance. He said **all points in space participate in the physical process, and the effect propagates from points to a neighboring points**. So in this principle of locality, you don't have action at a distance, action at a distance is always conveyed from one point to another point through the propagating in the space.

The concepts of **fields** is **the mathematical device or vehicle that the principle locality is at work**. The main idea of the field is that we associate each point in space a dynamical variable.

For example, if you have electric field is defined at each point x , we can introduce an electric field, which can also depend on time. Of course, normally we write it this way $\vec{E}_{\vec{x}}(t) \equiv \vec{E}(\vec{x}, t)$. The reason I write it in this way $\vec{E}_{\vec{x}}(t)$ to emphasize that in the definition of the electric field, the space and times actually play very different roles. The space plays the role of a label. So at each point $\vec{x}$, we have electric field, so x here is just a label, and t is used to describe the evolution, the change of the electric field. So $\vec{x}$ and t play very different roles.

Similarly, you can do this to magnetic field $\vec{B}_{\vec{x}}(t) \equiv \vec{B}(\vec{x}, t)$, you should always treat $\vec{x}$ as labels. And we know the evolution of electric and magnetic field is described by Maxwell's equations,

$$\begin{cases} \nabla \cdot \vec{B} = 0 \\ \nabla \cdot \vec{E} = 4\pi\rho \\ \frac{\partial \vec{B}}{\partial t} = -\nabla \times \vec{E} \\ \nabla \times \vec{B} = 4\pi\vec{J} + \frac{\partial \vec{E}}{\partial t} \end{cases} \quad (1)$$

which is called the differential form of the Maxwell's equations. This set of equations exemplifies perfectly the principle of locality. It's because you see those equations only involve the value of electric fields and the magnetic field at a single point. If $\vec{B}$ is at $\vec{x}$ point, then $\vec{E}$ is also at $\vec{x}$ point. You never say $\vec{B}_{\vec{x}}$ but $\vec{E}_{\vec{y}}$. So this

reflects the principle of locality that the effect is propagating from point to point, you just have the derivative of the same point. You don't involve separate points. So we call Maxwell's equations as local equations, they contain only E and B at same point with finite number of derivatives. So the derivatives are the ones to help you to propagate because it relates the point to the neighboring point.

Another example of locality is the Einstein's general relativity. In Einstein gravity, the dynamical variables are so-called spacetime metric $g_{\mu\nu}(\vec{x}, t)$. They are object with two indices, two spacetime indices, and the Einstein's equations are equations for this kind of object $g_{\mu\nu}$. And again this equation are local equations, in the sense that they only depends on the g evaluated at the same point with a finite number of derivatives. So in Einstein's gravity, you no longer have action at distance, so the effect of the gravity is propagated through the spacetime.

Here we say these two examples exemplify the principle of locality. In fact, the principle of locality plays a very important role in formulating those equations, which is a very powerful principle.

Lecture 2: Classical Field Theories

Types of field

- A scalar field $\phi(\vec{x}, t)$. This is a quantity defined at each point; there is only one value at the given point. For example, the temperature is a single scalar quantity defined at each point.
- A vector field: $\vec{E}(t, \vec{x})$ and $\vec{B}(t, \vec{x})$ are examples of vector fields because you have a vector at each point. Here, I'm using the three-dimensional notations, but in the relativistic theory, we can use the four vector notations, e.x., the vector potential has the form $A_\mu(x^\nu)$. Here A_μ is a four-vector, obtained by combining the spatial and temporal components into a single spacetime vector.
- A tensor field: $g_{\mu\nu}(x^\lambda)$ is an example for tensor field. Here you have two indices, you have many different components depend on the dimension. At each point, you have an object with two indices.
- Spinor fields $\psi_\alpha(t, \vec{x})$, where α is another index that we will define later.

Convention

- μ is always from 0 to 3, and sometimes I write simply x ; when I write $x = x^\mu = (ct, \vec{x}) = (ct, x^i)$, it denotes a four-vector. The spatial index is always denoted by i .
- $\partial_\mu = (\frac{1}{c}\partial_t, \nabla)$, ∇ is the gradient on the spatial directions.
- A four-vector $A_\mu = (A_0, \vec{A}) = (A_0, A_i)$, we still use i to denote the spatial components.
- $c, \hbar = 1$.

Action Principle in Classical Mechanics

First, recall in your classical mechanics, we introduce the action

$$S = \int dt L(x, \dot{x}, t), \quad (2)$$

where Lagrangian is a function of x , $\dot{x}$ and the time. This is the one-dimensional particle motion. And you can also introduce the canonical momentum, which is defined by

$$p = \frac{\partial L}{\partial \dot{x}}. \quad (3)$$

The Hamiltonian is related to the Lagrangian by

$$H = p\dot{x} - L, \quad (4)$$

through a Legendre transformation. And the equation of motion is obtained by extremizing $S[x(t)]$, where S can be consider as a functional of your trajectory, so whatever trajectory you have, you extremize this S , and you obtain the equation of motion (EOM).

Classical Field Theory

Now we can generalize this to field theory. From the principle of locality, the form of the Lagrangian L in field theory is significantly constrained. It must have the form:

$$L = \int d^3x \mathcal{L}(\phi_a, \partial_t \phi_a, \partial_i \phi_a) \quad (5)$$

which is the integration over all spatial directions. Here, $\phi_a(\vec{x}, t)$ is a function of spatial direction and time, it is a general field, and a label different fields. e.g., a can label different scalar fields if you have different scalar fields, a can also refer to A_μ or to spinor fields, and so on. $\mathcal{L}$ is called the Lagrangian density, which depends only on the value of ϕ_a and its derivatives at a single point. They are still local.

Remarks: Why the Lagrangian must have this form?

- First, the principle of locality implies (5) only involves a single integral because the locality does not allow something like this,

$$L = \cdots + \int d^3x d^3y \phi(\vec{x}, t) \phi(\vec{y}, t) \quad (6)$$

which involving the ϕ at different points. If your Lagrangian contains such terms, then when we derive the equations of motion, they will not be local. EOM will involve the behavior of your field at one point and then influenced by the field value at a distant point, then you will not have local equation.

- We only allow first derivative in time, we don't allow high order derivatives in time. The reason is that again the equation of motion will involve more than two derivatives in time if your Lagrangian has high order derivatives in time. $\partial_t \phi$ in (5) implies that EOMs contain two t -derivatives, which come from our experiences. In real life, all experiments are determined by the initial conditions. For the initial condition, you only need to specify the position and velocity; no higher derivatives are required. If the equation of motion involves more than two derivatives in time, then you need to specify more general initial conditions. But for simplicity, for the most of the time, as we will see, we will restrict to quantum field theory in first derivative in time.

So now as in classical mechanics, we can introduce the canonical momentum, Hamiltonian, and so on. Here, we can introduce the so-called canonical momentum density

$$\Pi_a(\vec{x}, t) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a(\vec{x}, t)}, \quad \text{where} \quad \dot{\phi}_a(\vec{x}, t) = \partial_t \phi_a \quad (7)$$

So this is just the direct generalization of classical mechanics. The momentum density is evaluated at each point, so Π_a also depends on $\vec{x}$ and t . The reason we call this the momentum density is because here $\mathcal{L}$ is the Lagrangian density. In the same way, we can also define the Hamiltonian density associated with each degree of freedom (DOF):

$$\mathcal{H} = \Pi_a \dot{\phi}_a - \mathcal{L} \quad (8)$$

Then, the Hamiltonian you just integrate over all space

$$H = \int d^3x \mathcal{H} \quad (9)$$

essentially a sum over all degrees of freedom. So now we can talk about equation of motion. It is analogous to classical mechanics. The classical field theory is just like classical mechanics, you generalize it to infinite number of DOFs. So now with each point in space, you associate it with some DOFs. Again, you just do the variation of the action, you extremize the action to be 0,

$$\delta S = 0 \quad (10)$$

where the action is again defined by

$$S = \int dt L = \int d^4x \mathcal{L}(\phi_a, \partial_\mu \phi_a). \quad (11)$$

You can straightforwardly generalize this to include higher spatial derivatives. Now let us compute the variation:

$$0 = \delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \delta (\partial_\mu \phi_a) \right] = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \partial_\mu (\delta \phi_a) \right] \quad (12)$$

In this case, δ is the variation, and ∂_μ and δ are independent, so you can exchange them, i.e.,

$$\partial_\mu (\delta \phi_a) = \delta (\partial_\mu \phi_a) \quad (13)$$

Then, we can do the integration by part for the second term and obtain

$$0 = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) \delta \phi_a \right] + \text{boundary terms} \quad (14)$$

The boundary terms will be proportional to $\delta \phi_a$. We will always choose $\delta \phi_a$ such that boundary terms vanish. Equation (14) must vanish for arbitrary variations $\delta \phi_a(x^\mu)$, which can be any function of x^μ . Therefore, the prefactor has to vanish, i.e.,

$$0 = \frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right), \quad (15)$$

(15) is the general EOM for classical field theory.

Restrictions in Field Theory

From now on, we will make two restrictions for simplicity.

Translationally invariant

The first is that we restrict to field theory which are translationally invariant, which means the theory does not single out any special point in spacetime. If you perform the experiment here in Chengdu, it is the same as performing it in Washington DC. If the theory were not translationally invariant, experiments performed in Chengdu and Washington DC would have different results.

Lorentz invariant

The second is that we assume the field theory is Lorentz invariant. A Lorentz invariant quantity looks exactly the same for every observer moving at constant velocity, no matter how their coordinate axes are rotated or how fast they move relative to each other. A quantity is Lorentz invariant if it does not change under any Lorentz transformation that includes both: Spatial rotations (changing direction in space), and Lorentz boosts (changing velocity or reference frame).

Now let us look at some simple examples of classical field theories that satisfy these two conditions.

E.x. 1: Maxwell's theory.

The dynamical variable is the four-vector potential $A_\mu(x^\nu)$. For A_μ , you define the field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (16)$$

The electric and magnetic fields E and B can be obtained from $F_{\mu\nu}$. The action for E&M can be obtained by

$$S_{EM} = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) \quad (17)$$

This is the action for electromagnetism in the absence of charged matter. If you do the variation here to get the EOM, then you obtain the source-free (vacuum) Maxwell equations. This action is corresponding to a local theory due to a single integral of spacetime.

Lecture 3: Symmetry and Conservation Law

Ex 2: Einstein gravity Einstein gravity is another classical theory. The Einstein-Hilbert action is

$$S = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} R, \quad (18)$$

where g is the determinant of the metric and R is the Ricci scalar. Einstein gravity is a local theory involving a single spacetime integral over geometric quantities.

Ex 3: The simplest quantum field theory the scalar field theory: A_μ is the vector field in E&M, and in Einstein gravity the dynamical variable is the metric tensor $g_{\mu\nu}$, which is even more complicated. Thus, the simplest case to study is a scalar field.

Let us just consider the real-valued scalar field $\phi(x)$. At each spacetime point, the field takes a single value; thus $\phi(x)$ is the dynamical variable. In physics, there are many examples of scalar fields; e.g., the Higgs field or the pion field, whose excitations correspond to scalar particles.

The simplest family of scalar field actions has the following form:

$$S = \int d^4x \left[-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right], \quad (19)$$

which is compatible with locality, translational symmetry, and Lorentz symmetry. Here, you can see that the Lorentz indices are contracted. The Lorentz indices in the derivative are contracted, so this guarantees the theory Lorentz invariant. This is essentially the simplest quantum theory one can write down based on locality.

In general, we can take $V(\phi)$ to be a polynomial

$$V(\phi) = \text{function of } \phi \quad (20)$$

When a system has translational symmetry, it means that if you perform an experiment anywhere, the result will be the same. This implies that all parameters in $V(\phi)$ must be constant and cannot depend on spacetime. The canonical momentum is

$$\Pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi} \quad (21)$$

where $\dot{\phi}^2 = \partial_t^2 \phi$ due to

$$\partial_\mu \phi \partial^\mu \phi = -\dot{\phi}^2 + (\nabla \phi)^2 \quad (22)$$

And then the Hamiltonian density is

$$\mathcal{H} = \Pi \dot{\phi} - \mathcal{L} = \frac{\Pi^2}{2} + \frac{1}{2} (\nabla \phi)^2 + V(\phi), \quad (23)$$

which is expressed in terms of momentum. And EOM is given by

$$\partial^2 \phi - \frac{\partial V}{\partial \phi} = 0. \quad (24)$$

and again I'm using a shorthand notation,

$$\partial^2 = \partial_\mu \partial^\mu = -\partial_t^2 + \nabla^2. \quad (25)$$

Here, let me give some simple examples of the V . In the simplest case, take $V = f\phi$. From translational invariance, f must be constant and cannot depend on spacetime. Then, the EOM is

$$\partial^2 \phi = f, \quad (26)$$

In this case, essentially you have some kind of constant, which is source for this ϕ . In this case, f acts as a constant source for the field ϕ . We say the system has an external force f ; because if $f \neq 0$, then ϕ cannot vanish. In this sense, the field ϕ is always generated by f . However, we usually consider systems without external forces, where the field evolves on its own.

In that case, if we ignore external forces, the simplest situation is

$$V(\phi) = \frac{1}{2} m^2 \phi^2 \quad (27)$$

m has to be constant. In this case, you have the EOM:

$$\partial^2 \phi - m^2 \phi = 0, \quad (28)$$

This is the famous equation called the KleinGordon equation. And you can also consider more complicated field theories with $V(\phi)$ to be some higher polynomial or more complicated functions. But keep in mind, this square is special as then the EOM has linear term. However, when $V(\phi)$ includes terms beyond quadratic order, the equation becomes nonlinear, and the analysis becomes more complicated. Anyway, this would be the simplest theory we consider.

Symmetry

A **symmetry** is a transformation of the dynamical fields that leaves the action invariant. Formally, each field ϕ_a transforms into a function of all the fields:

$$\phi_a \longrightarrow \phi'_a(\phi_b), \quad (29)$$

and if the action S remains unchanged under this transformation, we say the theory has a symmetry.

Example: Translation symmetry. Consider the scalar field theory introduced in (19). If we perform a constant shift of spacetime coordinates,

$$x^\mu \longrightarrow x'^\mu = x^\mu + a^\mu, \quad (30)$$

where a^μ is a constant four-vector, the scalar field transforms as

$$\phi(x) = \phi'(x'), \quad (31)$$

meaning that the field value at a given spacetime point does not change, only its coordinate label does. You can verify that the scalar-field action is invariant under this translation, confirming that translation is indeed a symmetry.

Example: Lorentz symmetry. Similarly, a Lorentz transformation acts on spacetime as

$$x^\mu \longrightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu, \quad (32)$$

where $\Lambda^\mu{}_\nu$ is a constant Lorentz transformation matrix satisfying $\Lambda^T \eta \Lambda = \eta$. Under this transformation, the scalar field again obeys (31). The action remains invariant, so Lorentz invariance is another continuous spacetime symmetry.

Continuous and discrete symmetries. Transformations such as translations and Lorentz transformations depend on continuous parameters (a^μ and $\Lambda^\mu{}_\nu$), so they are called **continuous symmetries**. In four dimensions, a^μ has four continuous parameters, while $\Lambda^\mu{}_\nu$ contains six (three rotations and three boosts).

The same scalar-field action also admits **discrete symmetries**, for example:

$$\phi \longrightarrow -\phi, \quad (33)$$

$$\vec{x} \longrightarrow -\vec{x}, \quad (34)$$

corresponding to field inversion and spatial parity. Both leave the action invariant but involve no continuous parameters. Because performing either transformation twice returns the system to its original state, each corresponds to a $\mathbb{Z}_2$ symmetry.

Global and local symmetries. Symmetries are further classified as **global** or **local**:

- *Global symmetries* have spacetime-independent transformation parameters. Translations and Lorentz transformations are examples, since a^μ and $\Lambda^\mu{}_\nu$ are constants.
- *Local symmetries* have spacetime-dependent parameters. A well-known example is the gauge symmetry in electromagnetism, where the phase transformation depends on spacetime position.

Noethers theorem. A deep connection exists between continuous symmetries and conservation laws: each continuous global symmetry corresponds to a conserved quantity. In classical mechanics, time translation implies energy conservation, spatial translation implies momentum conservation, and rotational symmetry implies angular momentum conservation. This general relationship is formalized by **Noethers theorem**.

Consider a continuous symmetry with an infinitesimal parameter ϵ :

$$\phi_a \longrightarrow \phi'_a = \phi_a + \epsilon f_a(\phi_b, \partial_\mu \phi_b), \quad (35)$$

where f_a is a function of the fields and possibly their derivatives. The variation of the Lagrangian density under this infinitesimal transformation must be a total derivative:

$$\delta \mathcal{L} = \epsilon \partial_\mu K^\mu, \quad (36)$$

for some current K^μ . If $K^\mu = 0$, the Lagrangian is strictly invariant.

Using the Euler-Lagrange equations,

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0,$$

the variation of $\mathcal{L}$ can be written as

$$\hat{\delta} \mathcal{L} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \hat{\delta} \phi_a \right) = \epsilon \partial_\mu K^\mu. \quad (37)$$

Comparing both sides yields the conservation equation

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} f_a - K^\mu \right) = 0. \quad (38)$$

Defining the **Noether current**

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} f_a - K^\mu, \quad (39)$$

we obtain the conservation law

$$\partial_\mu J^\mu = 0. \quad (40)$$

Continuity equation and conserved charge. Writing $J^\mu = (J^0, \vec{J})$, the conservation law takes the familiar form

$$\partial_0 J^0 + \nabla \cdot \vec{J} = 0, \quad (41)$$

which expresses the local conservation of a density and its associated current. Integrating over all space gives

$$Q = \int d^3x J^0, \quad (42)$$

and differentiating (41) shows

$$\partial_0 Q = 0, \quad (43)$$

so the total charge Q is conserved. Equation (43) is the integrated form of the local continuity equation (41).

Remarks. This result is known as **Noethers first theorem**, valid when ϵ is spacetime independent (global symmetry). When the transformation parameter depends on spacetime, as in gauge theories, additional terms appear and the statement becomes **Noethers second theorem**, which relates local symmetries to redundancies among the equations of motion.

The key idea is that any finite continuous symmetry can be constructed by successively applying infinitesimal transformations. Hence, understanding the infinitesimal form of a symmetry is sufficient to determine its full finite structure.